Analyzing the Relationship Between Staff and Resident COVID-19 Vaccination Rates Across Healthcare Facilities

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During the COVID-19 pandemic, it became very clear how important vaccinations are for saving weak groups, especially people who live in nursing homes and other long-term care facilities. People who live in these places often already have health problems, and they are more likely to get sick, end up in the hospital, or even die from COVID-19. On the other hand, those who are close to residents and have been vaccinated themselves significantly influence either the risk of spread's increase or decrease.

Good public health policies and facility management plans depend on an understanding of how workers' and residents' immunization rates interact. Actions taken against staff could have a greater impact on the safety of residents if staff vaccination rates are closely related to resident immunization rates. Also, if the connection is weak, distinct strategies could be required to ensure that every resident gets vaccinated. I am investigating the correlation between the number of immunizations administered to staff and residents in various healthcare institutions. I am also investigating the relevance of any variations we discover and creating predictive models to better understand how staff vaccination behavior influences the outcomes for residents.

Null Hypothesis (H_0): There is no correlation between resident and staff vaccination rates across facilities. Staff vaccination rates and resident vaccination rates are independent of each other. Knowing how vaccinated the staff are does NOT tell you anything useful about how vaccinated the residents are. Any observed pattern between staff and resident vaccination is just random.

Alternative Hypothesis (H_1): There is a correlation between resident and staff vaccination rates across facilities. There is a real relationship between staff and resident vaccination rates. When staff vaccination rates go up, resident vaccination rates also tend to go up (positive correlation), or vice versa (negative correlation, but in your case positive). This relationship is not random — there's an actual association between the two groups.

For Meta data, I have:

1. CMS Certification Number (CCN): replaces the term Medicare Provider Number, Medicare Identification Number or OSCAR Number. The CCN is used to verify Medicare/Medicaid certification for survey and certification, assessment-related activities and communications. The RO assigns the CCN and maintains adequate controls. In short, it is a unique identifier assigned by the Centers for Medicare & Medicaid Services to each healthcare provider or facility. (Wikipedia, 2022)
2. State: The US state where the facility is located
3. Percent of residence who are up to date on their vaccines: The percentage of nursing home residents at the facility who are considered up to date with their COVID-19 vaccinations
4. Percent of staff who are up to date on their vaccines: The percentage of staff members working at the facility who are considered up to date with their COVID-19 vaccinations.
5. Date vaccination data last updated: The most recent date on which the vaccination data for that facility was reported or updated.

I thought the response variable supposed to be Percent of residents who are up to date on their COVID-19 vaccination. Because the vaccination rate among residents is important in places like nursing homes because it shows how well the most vulnerable people are being protected. And the predictor variable is Percent of staff who are up to date on their COVID-19 vaccination. Because staff have close daily contact with residents, which can impact their health and facilities with high staff vaccination rates may enforce stricter policies, indirectly boosting resident vaccination rates. In this study, I investigated how staff and resident COVID-19 vaccination rates are related across healthcare facilities. The goal was to test the idea that staff vaccine rates are related to resident vaccination rates. Reader can understand the relationship between the rate of resident vaccination and the rate of staff vaccination directly and obviously.

When I was doing a correlation study, I found that the Pearson correlation coefficient was $r = 0.236$ and the p-value was close to zero (not equals zero). This means that there is a weak but statistically significant positive relationship between the vaccination rates of staff and residents. Then I reject the null hypothesis and accept the alternative hypothesis because of this. This proved that there is a relationship, even if it is not strong.

Using a z-test to test the hypothesis again showed a very high z-value ($z = 133$) and a p value that was almost zero. This showed that there was a very important difference between the average vaccine rates of residents (42.8%) and staff (7.6%). This finding shows that there is a big difference between these two groups, which makes it clear that they need different and more targeted vaccination plans.

A simple linear regression model was also made to predict the vaccine rates of residents based on the vaccination rates of staff. The model showed that there was a positive relationship, but the R^2 score was low. This means that even though staff vaccination rates do affect resident vaccination rates, they only explain a small part of the overall variation.

In conclusion, the analysis clearly shows that staff and resident vaccination rates are positively correlated, but not strongly. Residents are significantly more vaccinated than staff, and staff vaccination rates alone are not sufficient predictors of resident vaccination outcomes.

Future efforts to improve overall vaccination coverage should focus on independent strategies for residents and staff, and consider additional factors such as facility policies, public health interventions, and local COVID-19 risks. Also focus on expanding data collection and applying more advanced modeling techniques to better capture the complexity of vaccination behaviors across facilities.

During this project, I ran into a few problems with choosing datasets, cleaning data, writing, and figuring out what the results meant. It was very important to pick a good dataset because many of the ones that were available had missing or wrong data that needed to be carefully checked to make sure they were consistent and reliable. I dropped rows with empty values to keep the data correct, and I paid more attention to the data to make sure it wasn't wrong. When I was learning to code, I had trouble with libraries I wasn't familiar with. I fixed these problems by researching on Google and improving my method with the help of ChatGPT and Gemini. For graph visualization, I used templates from past labs to make clear plots. However, it was still hard to figure out what the weak trends were between staff and resident vaccination rates because heatmap number is positive but close to zero, too weak.

Reference

https://en.wikipedia.org/wiki/Centers_for_Medicare_%26_Medicaid_Services

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Source of data:

- https://data.cms.gov/provider-data/dataset/pvax-cv19
- https://data.cms.gov/provider-data/dataset/avax-cv19